MIL-D-11302D
13 February 1978
SUPERSEDING
MIL-D-11302C
24 November 1969

MILITARY SPECIFICATION

DERAIL, RAILWAY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a railway derail used as a safety measure at a siding or clearance to insure protection from intruding equipment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 2250

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The derail shall be either the hinged-block or sliding-block type of malleable iron, cast-steel, or fabricated steel construction as specified. The derail shall consist of derail block and base section. The derail shall be of rugged construction and shall be designed so that the block can be manually positioned on running rails of various sizes and locked in place or positioned on the base section to provide clearance for moving equipment. Provisions shall be made for attaching the derail to the rail or tie so that the derail shall not be dislodged or loosened upon impact from a railway car with a gross weight of 75 tons (68 tonnes) traveling at a minimum of 5 mph (8 kmph). The derail shall be for right- or left-hand rail mounting (see 6.3) and shall be satisfactory for use with the rail weight and height specified.

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3.2 First article.

3.2.1 First-produced derail. The contractor shall furnish a derail for examination and test within the time frame specified to prove that his production methods will produce derails that comply with the requirements of this specification (see 6.4). Examinations and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Material. Material shall be as specified herein (see 6.5).

3.4 Identification marking. The derail shall be identified in accordance with MIL-STD-130.

3.5 Workmanship. All parts of the derail shall be clean and free from sand, dirt, fins, pits, cracks, nicks, flaws, and seams.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced derail inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced derail inspection.

4.3.1 Examination. The first-produced derail shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. The first-produced derail shall be tested as specified in 4.5.2. Failure to pass any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. The derail shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The derail shall be tested as specified in 4.5.2.2. Failure to pass the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The derail shall be examined for the following defects:

101. The derail not furnished as specified (see 3.1).
102. Material not as specified (see 3.3).
103. Identification marking missing or illegible or not in accordance with MIL-STD-130 (see 3.4).
104. Workmanship not as specified (see 3.5).

4.5.2 Tests.

4.5.2.1 Impact test. The first-produced derail shall be attached to a rail or tie and shall be hit one time by a railway car having a gross weight of 75 tons (68 tonnes) traveling at a minimum speed of 5 mph (8 kmph) to determine conformance to 3.1.

4.5.2.2 Clearance test. The block shall be positioned manually on the base section and a railway car moved over it to determine that the clearance conforms to 3.1.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. The derail shall be examined for the following defects:

105. Packing not Level A, Level B, or Commercial as specified (see 5.1).
106. Derails not packed for Level A as specified (see 5.1.1).

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107. Derails not packed for Level B as specified (see 5.1.2).
108. Derails not packed for Commercial level as specified (see 5.1.3).
109. Marking for military level of packaging not in accordance with MIL-STD-129 (see 5.2.1).
110. Marking for Commercial level of packaging not as specified (see 5.2.2).

5. PACKAGING

5.1 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Derails of like description shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.1.2 Level B. Derails shall be packed as specified in 5.1.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.1.3 Commercial. Derails shall be packed to assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. Marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The derail is intended for use as a safety device for derailing railroad rolling stock in case of an emergency.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and construction of derail (see 3.1).
(c) Whether derail is for right- or left-hand mounting (see 3.1 and 6.3).
(d) Weight and height of rail with which derail is to be used (see 3.1).
(e) Time frame required for submission of first-produced derails (see 3.2.1).
(f) Degree or packing required (5.1).
(g) When Finish A strapping is required (see 5.1.1).

6.3 Right- and left-hand derail. Looking in the direction of movement of a car to be derailed, a right-hand derail is placed on the right-hand side and derails toward the right. A left-hand derail is the converse of the above.

6.4 First-produced derail. Any changes or deviations of production derails from the approved first-produced derail during production will be subject to the approval of the contracting officer. Approval of the first-produced derail will not relieve the contractor of his obligation to furnish derails conforming to this specification.

6.5 Recycled material. It is encouraged that recycled materials be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
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Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-11302D Derail, Railway

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